



Introduction to the Deontology: Assigning Grades

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Student Materials

- web
- PDF

Overview

Ethics background required: Familiarity with Virtue Ethics and the need for the study of ethics in computer and data sciences is helpful

Subject matter referred to in this lab: algorithms, artificial intelligence, computer programming

Placement in overall ethics curriculum:

- Academic year: First year, second semester, probably second semester programming course
- Recommended previous labs: Virtue Ethics and/ or A First Look at Ethics in Computer and Data Science
- Recommended follow-up labs: Utilitarianism-Plagiarism Issues

Time required:

- Out of class: None
- In class: 25-30 minutes

Learning objectives:

- Students understand the utility of the deontological framework

Lab overview:

Ethical issue to be considered: (AI) decision making

Flow

- Students are reminded of what the virtue ethics framework is and discuss some of its challenges



- Students are presented with the idea of the deontological framework for choosing an ethical course of action

- Activity: Students have the opportunity to find an ethical solution to assigning a letter grade from a numerical one
- Reflection: Students share virtue ethics and deontological solutions to the problem of assigning a fair letter grade
- Students are introduced to the ACM code of ethics

Preparation required: Read the entire lab. Print a copy of the student handout for each student in the class or provide access to a document posted online. Have a copy of the professor “answers” available in case you need them.

Guide for Instructors

Lesson plan

Introduction (to be read or summarized to class) (5 minutes)

Recall that ethics is the study of determining the right thing to do (in a moral sense). In designing computer programs and performing data analysis, there are lots of these types of decisions to be made. You may have been introduced in previous labs of the virtue ethics framework to assist in making a decision that is morally “best.” (If students have seen the virtue ethics material, ask what they remember about it)

The virtue ethics framework asks the question “what would a virtuous person do” to determine what is right. A virtuous person, according to Aristotle, would have good underlying character traits that would cause them to make good choices. The main benefit of this framework is that it is very flexible. It can be applied to a wide variety of decisions.

What are some of the challenges of using the virtue ethics framework? (Let students answer, but here are 2 obvious problems)

- Finding a truly virtuous person to make decisions
- Some virtues may appear to be in conflict with each other (kindness and honesty, for example), making the decision less obvious. (How to answer someone who asks what you think of the book that they wrote – when you don’t like it)

Would it be easier to just have a set of rules to follow that will always produce the ethical action? This is the idea behind the deontological approach to ethical decision making.

According to Oxford Languages, Artificial Intelligence is “the theory and development of computer systems able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.”



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Any time you've written a computer program that uses "if statements" you're writing a program that makes decisions – that employs AI. What examples can you think of? (let students answer)

- Perhaps the students written programs determining GPA, whether a password is strong enough, the next best move in a game, which loan terms are better

Your program basically consisted of a set of rules (an algorithm) for making the decision. Perhaps the decisions your programs have made in the past have not seemed like ethical decisions. A commonly cited example of developing an ethical set of rules with AI is that of programming a driverless car to decide who/what to hit when a collision is inevitable. (The small group, the large group? The elderly person or the younger person? Save the passengers or the pedestrians?)

But we will consider something more familiar: the assignment of a letter grade. Most students have written a program to assign a letter grade given a numerical grade. Commonly, 93 and above is an A, 90 up to less than 93 is an A- etc. But how many times have you been close to a grade and felt that just going by the rubric was not "fair"?

Activity (7-10 minutes)

Best to put the students into groups of 3-4. Can be completed individually, however.

You will be asked to consider what letter grade should actually be assigned (based on the numerical grade) from the virtue ethics standpoint, and the deontological approach. When it comes to the rules-based deontological decision making process, you may just want to stick to the rubric (which already is a rule!), or you may want to take other things into consideration.

Reflection (to be completed in class after the activity is finished) (10-15 minutes)

Have class come back together. Each group should explain their virtue ethics decision (along with the virtues that influenced the choice).

Once all groups have voiced their virtue ethics conclusion, have each group provide their algorithm.

If students have had a hard time coming up with an algorithm, or if the algorithms are too simple, consider discussing the following things that might come into play (not saying they should!):

- Did the student have one exam or assignment that was uncharacteristically low?
- Is the student ready to graduate (and an F would keep them from graduation)
- Did the student have a personal trauma during the semester
- Is the professor confident that the student did their own work all semester, or is there evidence that was not the case



- How did the rest of the class do?

Discuss the different approaches (virtue ethics vs. deontological) and the challenges and benefits of each in this scenario.

- Deontological would provide a more uniform decision and is easier on the person making the decision.
- Virtue ethics is flexible – you do not have to try to come up with a rule for every possibility

Discuss also, the relationship between the 2 approaches.

- Students may observe that rules are related to virtues – compassion, for example, if a student had a particularly bad week

Coming up with a set of rules is often helpful, though challenging. Some companies or organizations come up with guidelines that are helpful, but that leave room for interpretation or special cases. The most famous is the American Medical Association's "do no harm." The Association for Computing Machinery (or ACM -the main Association for computing professionals) has a code of ethics. Part of this code of ethics is found on the second page (or backside) of the student handout. Let the students read and comment if there is time.

Assessment

1. Briefly describe the deontological approach to determining if something is the right course of action.
 - Following a set of rules or guidelines. If a course of action adheres to the rules, it is the right thing to do.
 2. What are the benefits of the deontological approach and what are the drawbacks
 - Benefit is that the result is more uniform and easier for the person deciding
 - Drawback is that it is hard to come up with a set of rules that covers every possibility
 3. Provide a scenario and ask student to come up with a set of rules to ethically decide the outcome.
 4. Ask student to identify a set of rules that society follows:
 - Traffic rules
 - Ten Commandments
 - Classroom rules
 - Any set of governmental laws
- How is artificial intelligence related to the deontological approach?
 - Both follow a set of rules to make decisions



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- Name one article of the ACM Code of Ethics
 - There are many!

Bibliography